# SARIMA

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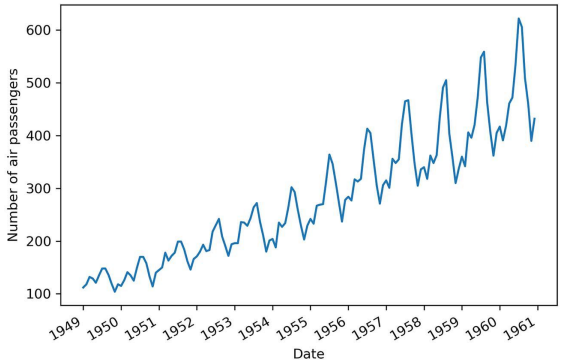
# Accounting For Seasonality

This chapter covers

- Examining the seasonal autoregressive integrated moving average model, SARIMA(p,d,q)(P,D,Q)m
- Analyzing seasonal patterns in a time series
- Forecasting using the SARIMA(p,d,q)(P,D,Q)m model

**ARIMA model**: It helps us forecast time series data (like sales or passenger counts) when the data isn't stable over time.
**SARIMA model**: This is just ARIMA with an extra twist — it adds seasonal components. That means it can handle repeating patterns that happen at regular intervals (like every year, every quarter, or every month).
**Airline passenger example**




The data shows a clear yearly pattern — fewer passengers in winter, more in summer.
ARIMA alone can't fully capture this repeating cycle.
SARIMA adds seasonal parameters so the model can learn both the overall trend **and** the seasonal ups and downs.

# Accounting For Seasonality

**SARIMA model** is just ARIMA with extra seasonal parts.
It adds four new parameters: **P, D, Q, and m**.

- **m** = the length of the seasonal cycle (e.g., 12 for monthly data in a year, 4 for quarterly data, 7 for weekly cycles in daily data).
- **P** = seasonal autoregressive part (uses past seasonal values, like last year's July).
- **D** = seasonal differencing (removes seasonal trends to make the series stable).
- **Q** = seasonal moving average part (uses past seasonal errors).

**Example with airline passengers:**

- Data is monthly, so **m = 12**.
- If **P = 2**, the model looks at passenger numbers from 12 and 24 months ago.
- If **D = 1**, it subtracts this year's value from last year's same month to remove seasonality.
- If **Q = 2**, it uses error terms from 12 and 24 months ago.

SARIMA learns from both the overall trend **and** the repeating seasonal cycle (like summer travel peaks) so forecasts are more accurate for data with regular patterns.

## Identifying seasonal patterns in a time series

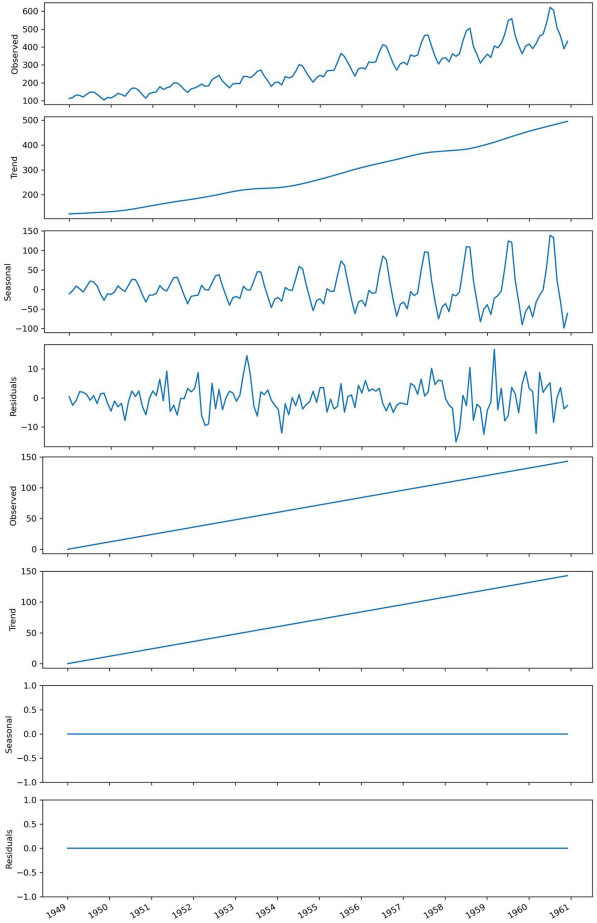
**Seasonality** means a pattern that repeats at regular intervals (like every year, every month, or every week).
To check if your data has seasonality, you can:

**Plot the data**: Often, just looking at the graph shows repeating ups and downs.
Example: Airline passengers rise every summer and drop in winter.

**Decompose the time series**: This splits the data into three parts:

- **Trend** → the long-term direction (e.g., overall growth in air travel).
- **Seasonal** → the repeating cycle (e.g., summer peaks, winter lows).
- **Residuals (noise)** → random fluctuations that don't follow a pattern.

# Accounting For Seasonality




The **observed plot** is the original data (what we actually see).
The **trend plot** shows the long-term movement (like steady growth).
The **seasonal plot** shows the repeating ups and downs (like summer peaks, winter lows).
The **residuals plot** shows the leftover random noise.

Each of these components has its own scale on the y-axis because they represent only part of the variation. When you **add them together (trend + seasonal + residuals)**, you get back the original observed data.

That's why the seasonal part sometimes goes negative or positive — it's just showing how much it pushes the data up or down to create the peaks and troughs.

Time series decomposition of a simulated linear series.
The **observed data** is  perfectly linear series.
the **trend component**, which is e same as the observed data, since the series is linearly increasing over time.
Since there is no seasonal pattern, the seasonal component is a flat horizontal line at 0
Residuals are also 0, because it is perfectly simulated linear time series

# Accounting For Seasonality

## Forecasting the number of monthly air passengers

**Collect the data** – same as before.
**Check for stationarity** – if the series isn't stable, apply transformations and differencing.

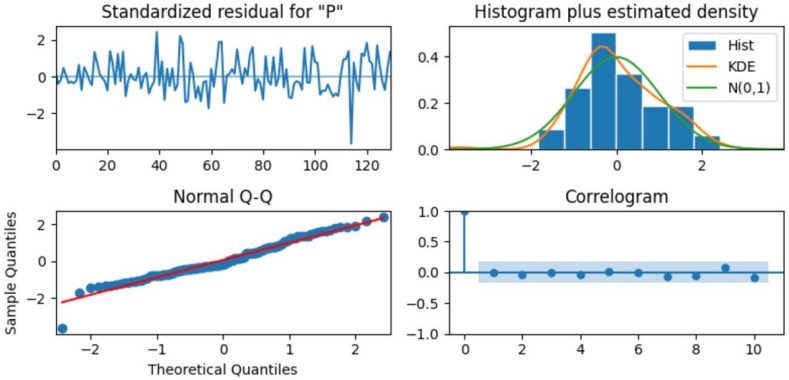
- **d** = number of times we difference to remove trend.
- **D** = number of times we apply *seasonal differencing* to remove repeating cycles.

**Set ranges for parameters** – now we have both regular (p, q) and seasonal (P, Q) autoregressive/moving average terms.
This makes the search space bigger, so testing models takes longer.
**Fit multiple models** – try different combinations of (p,d,q)(P,D,Q)m.
**Select the best model** – choose the one with the lowest AIC (a measure of model quality).
**Check residuals** – make sure the leftover noise looks random.
**Forecast** – once the model is validated, use it to predict future

**Example with airline passengers:**

- Data is monthly, so **m = 12**.
- We want to forecast **12 months ahead (1961)**.
- Compare a simple baseline (naive seasonal forecast) with ARIMA and SARIMA.
- SARIMA should perform better because it captures the yearly seasonal cycle (summer peaks, winter lows).

## Forecasting using ARIMA



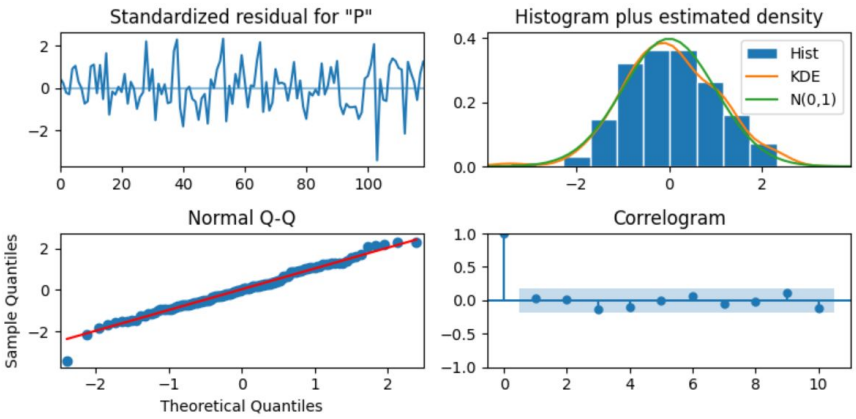

he residuals behave like white noise, which is good. It means the ARIMA(11,2,3) model has captured most of the structure in the data, leaving only random noise behind.

## ljung box test result

The ARIMA model isn't perfect here — it misses some seasonal structure. That's why SARIMA, which handles seasonality, is the better choice for forecasting airline passengers.

# Accounting For Seasonality

## Forecasting using SARIMA

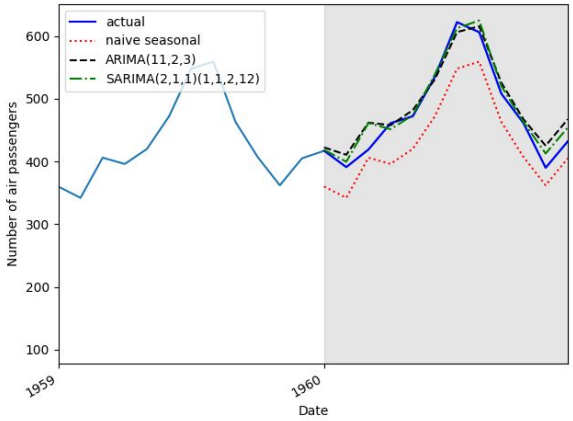


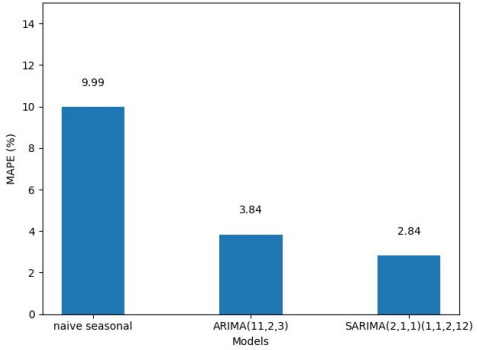

The residuals behave like white noise, which means the SARIMA(2,1,1)(1,1,2)$_{12}$ model has captured the seasonal structure well and is a good fit for the data.

### ljung box test result

The returned p-values are all greater than 0.05. Therefore, we do not reject the null hypothesis, and we conclude that the residuals are independent and uncorrelated, just like white noise

## Compare Results




**SARIMA has the lowest MAPE**

# Accounting For Seasonality

## Next

**SARIMA** adds seasonal parameters (P, D, Q, m) to ARIMA so we can model repeating cycles in time series.
**P** = seasonal autoregressive order (past seasonal values),
**D** = seasonal differencing (removes seasonal trend),
**Q** = seasonal moving average order (past seasonal errors),

- **m** = frequency of the cycle (e.g., 12 for monthly data in a year).
- We learned how to spot seasonality using **plots** and **time series decomposition** (trend + seasonality + noise).
- Step by step, we built models: MA → AR → ARMA → ARIMA → SARIMA.
- But SARIMA only uses the time series itself.

Now comes the **SARIMAX model**:

- The **X** stands for **exogenous variables** (external factors).
- This means we can include outside influences (like interest rates, debt levels, holidays, or weather) along with the time series to make better forecasts.